MOBILE APPLICATION DEVELOPMENT — TERM PROJECT REPORT

Medicine Reminder: An Intelligent Clinical Dosage, Circadian Timing, Family Caregiver & Medication History System

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COURSE
Mobile Application Development / Mobile Programming

PLATFORM & ARCHITECTURE
Flutter (Dart) & Cross-Platform Mobile

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1. EXECUTIVE SUMMARY & CLINICAL ENGINEERING RATIONALE

Traditional medication reminder applications often suffer from a fundamental oversimplification: they treat every medication as an isolated alarm that sounds at the same hour every day. However, clinical pharmacology and real-world patient workflows reveal four critical challenges:

- **Non-Daily Treatment Intervals:** Many therapeutic regimens (e.g., specific antibiotics, immunosuppressants, corticosteroids, and high-dose vitamin boosters) are prescribed to be taken **every 2 days, every 3 days, every 4 days, or weekly**, rather than daily.
- **Multi-Dose Regimens & Independent Alarms:** Many essential medications (such as analgesics like Paracetamol, antibiotics, or cardioprotective drugs) must be taken multiple times per day (e.g., **Morning and Evening**, or **Morning, Noon, and Evening**). Each dose requires its own independent circadian alarm time, while drawing from a single, unified physical medicine stock.
- **24/7 Continuous Daily Auto-Reset:** For recurring daily therapies (`freq = 1`), alarms must automatically re-arm every single day. The system must automatically detect the change of date at midnight or upon morning launch and reset the active cycle state (`isTaken = 0`) without requiring manual re-entry.
- **Medication Recall & Double-Dosing Hazards:** Patients frequently ask themselves: *"Did I take my medicine yesterday? At what time and which dose did I take?"* Forgetting past doses leads to accidental double-dosing or missed therapeutic windows.
- **Family Caregiver Requirements & Profile Lifecycle:** Parents managing treatments for children or adults caring for elderly parents need to track multiple treatment regimens from a single device, complete with the ability to safely remove secondary profiles with cascade cleanup when therapy ends.
- **Formulation-Specific Safety Precautions:** Ophthalmic drops, nasal decongestants, asthma inhalers, and injections require strict clinical safety guidelines (such as the 5-day decongestant rebound congestion rule or rinsing mouth after steroid inhalers to prevent oral thrush).

To address these clinical realities, **Medicine Reminder** was engineered as an end-to-end, safety-oriented personal health and therapy assistant. It integrates a **7-formulation clinical engine** (Tablets, Syrups, Injections,

Drops, Sprays, Topical Creams, and Wound Dressings), an **Early Dose Guardrail** that prevents premature drug administration, card-level in-place plan editing, a user-configurable interval scheduling algorithm, a bidirectional double-table SQLite medication history journal, context-aware meal timing, multi-profile family caregiver segregation, real-time inventory tracking, and full bilingual internationalization (English & Turkish).

2. SYSTEM ARCHITECTURE & DIRECTORY HIERARCHY

```
Medicine_Reminder/
├── lib/
│   ├── main.dart                     # Application bootstrap, Material 3 theming & notification init
│   ├── models/
│   │   └── pill_model.dart           # Pill domain model (id, profileId, name, dosage, timeOfDay, freq, sto
│   ├── services/
│   │   ├── db_helper.dart            # SQLite double-table persistence & history logging
│   │   └── notification_service.dart  # Timezone-aware local background notification manager
│   └── screens/
│       ├── home_screen.dart          # Three-tab dashboard (Today / All Plans / History Log), profile switc
│       └── add_pill_screen.dart      # Form-factor selectors, dynamic dosage chips, interval stepper & clin
├── preview/
│   └── index.html                    # Responsive mobile web preview with Web Audio API synthesizer
├── screenshots/                      # High-resolution un-clipped native application screenshots
└── pubspec.yaml                      # Dependency configurations & SDK constraints
```

Core Technology Stack

- **Flutter & Dart SDK (v3.0+):** Declarative, reactive UI rendering with 60/120 FPS high performance.
- **SQLite (`sqflite`):** Embedded relational database engine supporting atomic transactions, foreign keys, and indexed queries.
- **Local Notifications (`flutter_local_notifications` + `timezone`):** Background operating system alarms that trigger sound and vibration even when the app is terminated.
- **Web Audio API Synthesizer (Preview):** Zero-latency multi-frequency chime engine designed for cross-platform compatibility on modern mobile browsers.
- **Bilingual i18n Engine:** Instant runtime switching between English (GB) and Turkish (TR) without application restarts.

3. RELATIONAL DATABASE SCHEMA

The persistence layer utilizes two relational tables in SQLite to separate scheduled therapy plans from historical audit logs.

Table 1: `pills` (Active Medication Plans)

COLUMN	TYPE	CONSTRAINT	ENGINEERING PURPOSE
`id`	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique identifier used for alarm notification IDs
`profileId`	TEXT	DEFAULT 'p1'	Caregiver family profile assignment (`p1` , `p2` , `p3`)
`name`	TEXT	NOT NULL	Generic or brand name of the pharmaceutical
`type`	TEXT	DEFAULT 'tablet'	Formulation: `tablet` , `syrup` , `injection` , `drops` , `spray` , `cream` , `dressing`
`dosage`	TEXT	NOT NULL	Complete dosage and administration route
`time`	TEXT	NOT NULL	Scheduled reminder time in 24-hour format (`HH:mm`)
`timeOfDay`	TEXT	DEFAULT 'morning'	Circadian interval (`morning` , `noon` , `evening` , `night`)
`freq`	INTEGER	DEFAULT 1	Interval period in days (**1: Daily**, **2: Every 2 days**, **X: Every X days**)
`stock`	INTEGER	DEFAULT 30	Remaining physical inventory; decrements on confirmation
`groupId`	TEXT	NULLABLE	Unique group identifier linking multi-dose entries (e.g. Morning & Evening doses of same drug) for shared inventory tracking
`lastTakenDate`	TEXT	NULLABLE	ISO-8601 date string (`YYYY-MM-DD`) of the most recent confirmed dose, enabling next-day automatic reset
`nextDueDate`	TEXT	NOT NULL	Target calendar date in ISO-8601 (`YYYY-MM-DD`)
`isTaken`	INTEGER	DEFAULT 0	Status for the current active cycle (0: Pending, 1: Taken)

Table 2: `history_logs` (Immutable Medication Logbook)

COLUMN	TYPE	CONSTRAINT	ENGINEERING PURPOSE
`id`	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique log entry identifier
`pillId`	INTEGER	FOREIGN KEY (`pills.id`)	Relational link to the prescribed medication
`profileId`	TEXT	DEFAULT 'p1'	Profile identifier for filtering historical family records
`name`	TEXT	NOT NULL	Preserved medication name at time of administration
`dosage`	TEXT	NOT NULL	Exact dosage taken
`time`	TEXT	NOT NULL	System timestamp when user tapped the confirmation button
`timeOfDay`	TEXT	DEFAULT 'morning'	Circadian interval marker
`dateYMD`	TEXT	NOT NULL	ISO date stamp (`YYYY-MM-DD`)
`dateLabel`	TEXT	NOT NULL	Human-readable relative date (`Today` , `Yesterday` , `2 Days Ago`)

4. VISUAL APPLICATION WALKTHROUGH & PRODUCTION SCREENSHOTS

The figures below provide full-resolution, authentic production captures directly from the latest system deployment. Each view highlights critical clinical workflows—including consolidated household adherence tracking, multi-profile caregiver management with clinical demographics, comprehensive treatment creation, safety guardrails during regimen modification, immutable chronological administration logs, and physician consultation summaries:

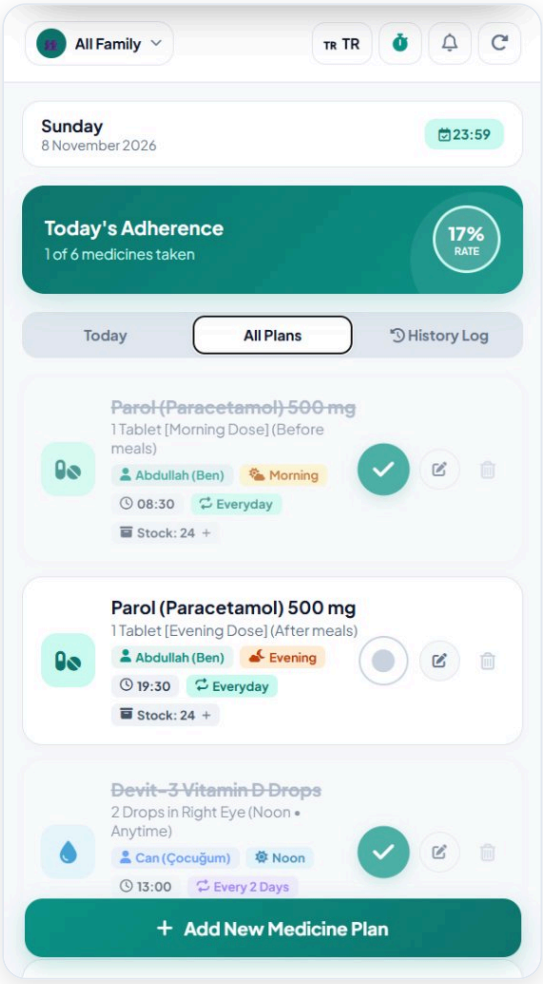


Figure 4.1: Consolidated Family Dashboard & All Plans Management

Illustrates the primary caregiving command center under the **"All Plans"** view with "*All Family*" selected. The real-time adherence header quantifies household compliance (17% completed, 1 of 6 doses taken). Each card distinctly exhibits the patient ownership chip (e.g. *Abdullah [Ben]*, *Can [Çocuğum]*), circadian dose slot (*Morning*, *Evening*, *Noon*), scheduled alarm time, frequency cycle (*Everyday*, *Every 2 Days*), remaining stock level with rapid refill button (+), interactive completion checkmark, in-place edit action ( ), and safe plan removal ( ).

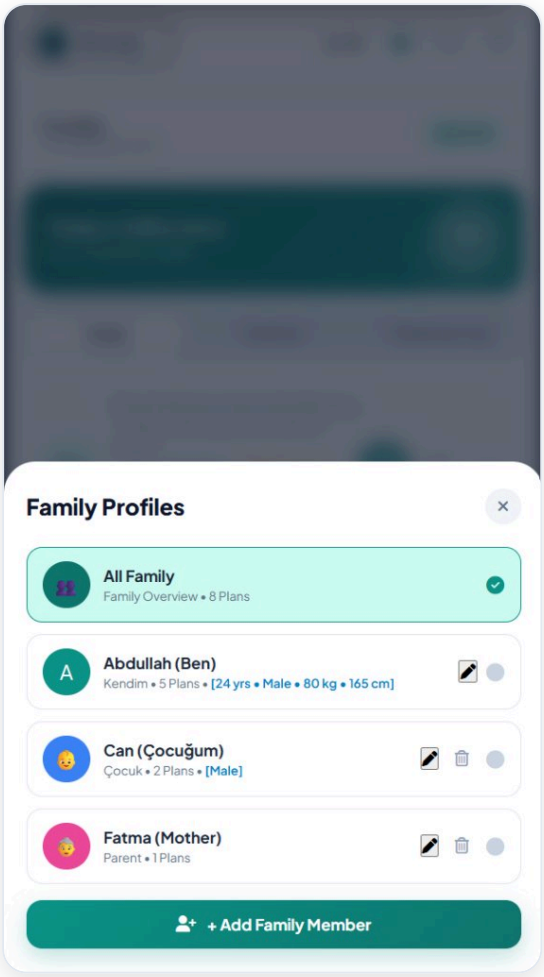


Figure 4.2: Family Caregiver Profiles Sheet with Clinical Demographics & Lifecycle Actions

Enables comprehensive multi-user caregiving from a single device. Caregivers can switch between "*All Family*" (consolidated household view of 8 plans), "*Abdullah (Ben)*" (5 plans, displaying clinical demographic badge **[24 yrs • Male • 80 kg • 165 cm]**), "*Can (Çocuğum)*" (2 plans, displaying **[Male]**), and "*Fatma (Mother)*" (1 plan). Each secondary profile provides dedicated 🖍️ **Edit** (to update name, relation, age, weight, and height) and 🗑️ **Cascade Delete** buttons, while the primary account is safeguarded from accidental deletion. The prominent **"+ Add Family Member"** button enables friction-free creation of additional care recipients.

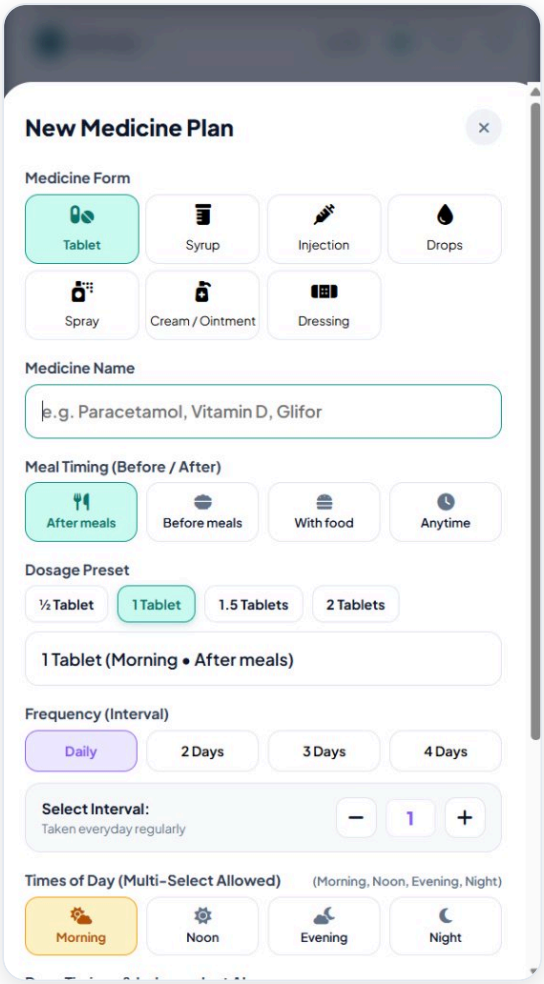


Figure 4.3: New Medicine Plan Form with 7 Formulations & Multi-Dose Timings

Showcases the comprehensive treatment creation interface supporting **7 clinical formulations** (Tablet, Syrup, Injection, Drops, Spray, Topical Cream/Ointment, and Wound Dressing). Features context-aware meal intake chips (*After meals*, *Before meals*, *With food*, *Anytime*), standardized dosage chips (*½ Tablet*, *1 Tablet*, *1.5 Tablets*, *2 Tablets*), real-time instruction preview, custom interval stepper (*[-] 1 [+]*), and simultaneous multi-dose circadian selection (*Morning*, *Noon*, *Evening*, *Night*) synchronizing independent alarm clocks to a single physical inventory.

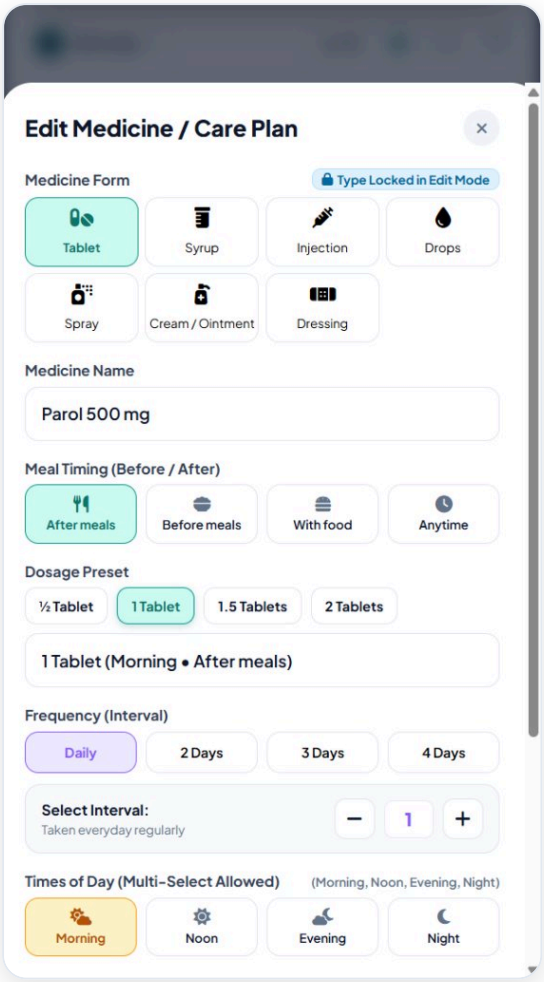


Figure 4.4: Edit Medicine / Care Plan Modal with Clinical Formulation Locking

Highlights the clinical safety guardrail during regimen modification. When editing an active treatment plan (e.g. *Parol 500 mg*), the system displays the authoritative **" 🔒 Type Locked in Edit Mode"** badge. Tapping alternative formulations is intercepted to prevent accidental cross-formulation mutation (e.g. converting an oral tablet into ophthalmic drops). Caregivers can safely adjust dosages, instructions, alarm times, and repeat frequencies while preserving pharmaceutical integrity.

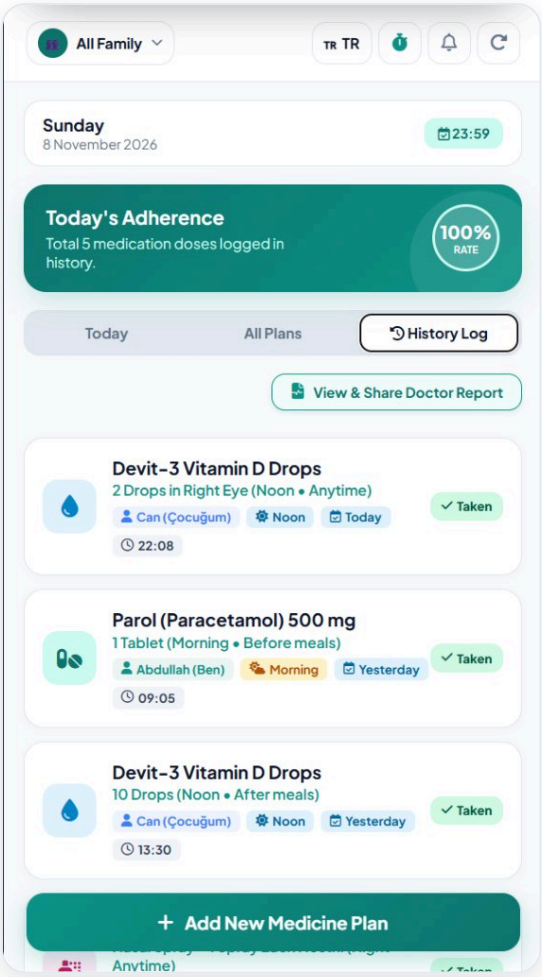


Figure 4.5: Chronological Medication History Log & Doctor Report Action

Presents the audit-grade, chronological administration log with 100% adherence rate across 5 recorded doses. Prominently features the green **"View & Share Doctor Report"** trigger. Each historical entry details the recipient patient, formulation icon, exact dosage instruction, circadian slot badge (*Noon*, *Morning*), calendar date (*Today*, *Yesterday*), verified administration timestamp (*22:08*, *09:05*, *13:30*), and a verified green status badge. This log eliminates double-dosing risks and provides verifiable compliance data for medical consultations.

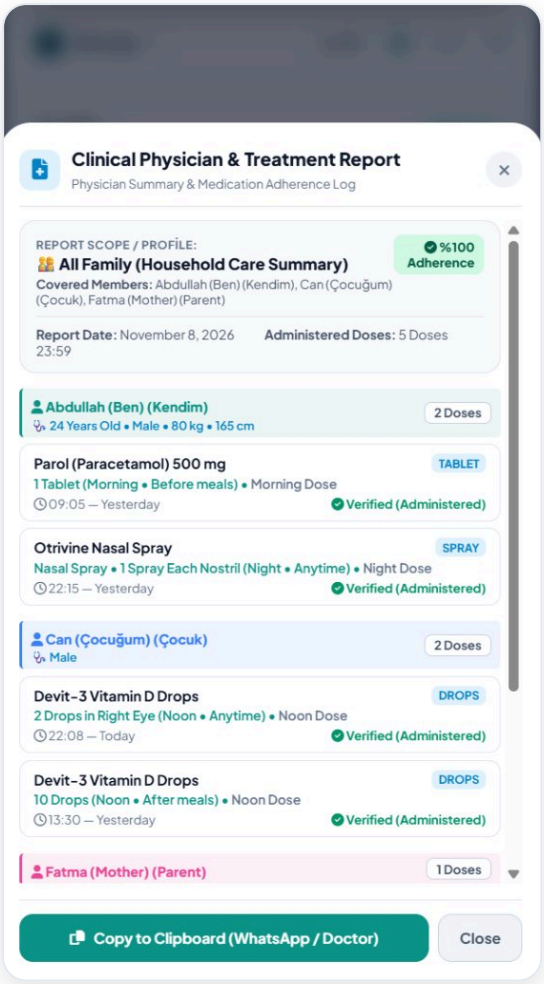


Figure 4.6: Interactive Clinical Physician & Treatment Report Modal with Patient Demographics

Demonstrates the doctor-facing consultation modal. Provides a complete household summary (100% Adherence, 5 Doses) with an itemized patient-by-patient breakdown. Crucially displays clinical demographic parameters (e.g. *Abdullah [Ben]: 24 Years Old • Male • 80 kg • 165 cm* and *Can [Çocuğum]: Male*), providing physicians with verified body weight and age metrics essential for pediatric and geriatric mg/kg dosage verification. Each medication entry displays formulation badges (*TABLET*, *SPRAY*, *DROPS*), meal timing, administration timestamps, and a one-click **"Copy to Clipboard (WhatsApp / Doctor)"** export action.

5. KEY ENGINEERING IMPLEMENTATIONS & SOURCE CODE

5.1. Multi-Dose Shared Inventory Depletion & Daily Frequency Logic

When multiple daily doses are scheduled for a single medication (e.g. Parol Morning at 08:30 and Evening at 19:30), they are assigned a shared `groupId`. Taking any dose synchronizes physical stock depletion across all linked records in the group while advancing the individual dose's history log:

```dart
// lib/services/db_helper.dart
Future<void> markPillAsTaken(Pill pill) async {
  final db = await database;
  final now = DateTime.now();
  final timeStr = "${now.hour.toString().padLeft(2, '0')}:${now.minute.toString().padLeft(2, '0')}";
  final dateYMD = "${now.year}-${now.month.toString().padLeft(2, '0')}-${now.day.toString().padLeft(2, '0

  await db.transaction((txn) async {
    final newStock = pill.stock > 0 ? pill.stock - 1 : 0;
    String nextDate = dateYMD;
    if (pill.freq > 1) {
      final nextDateTime = now.add(Duration(days: pill.freq));
      nextDate = "${nextDateTime.year}-${nextDateTime.month.toString().padLeft(2, '0')}-${nextDateTime.da
    }

    // 1. Synchronize shared physical stock across all multi-dose records in this group
    if (pill.groupId != null && pill.groupId!.isNotEmpty) {
      await txn.update(
        'pills',
        {'stock': newStock},
        where: 'groupId = ?',
        whereArgs: [pill.groupId],
      );
    }

    // 2. Mark this specific circadian dose as taken and set lastTakenDate
    await txn.update(
      'pills',
      {
        'isTaken': 1,
        'stock': newStock,
        'lastTakenDate': dateYMD,
        'nextDueDate': nextDate,
      },
      where: 'id = ?',
      whereArgs: [pill.id],
    );

    // 3. Record entry in immutable history logbook
    await txn.insert('history_logs', {
      'pillId': pill.id,
      'profileId': pill.profileId,
      'name': pill.name,
      'dosage': pill.dosage,
      'time': timeStr,
      'timeOfDay': pill.timeOfDay,
      'dateYMD': dateYMD,
      'dateLabel': 'Today',
    });
  });
}
```

5.2. Next-Day Auto-Reset & 24/7 Recurring Alarm Engine

To ensure daily medications (`freq = 1`) reliably trigger alarms every single day without manual reset, the application executes a continuous date-comparison check against `lastTakenDate`. When a new day begins, past-day completions are automatically reset to pending:

```
// preview/index.html & lib/services/db_helper.dart
function checkAndResetDailyPills() {
  const todayYMD = getTodayYMD(); // "YYYY-MM-DD"
  let changed = false;

  pills.forEach(p => {
    // For daily medicines (freq === 1), if last taken on a previous day, auto-reset for today
    if (p.freq === 1 && p.isTaken === 1) {
      if (!p.lastTakenDate || p.lastTakenDate < todayYMD) {
        p.isTaken = 0; // Re-arms alarms and resets checkmark for the new day
        p.nextDueDate = todayYMD;
        changed = true;
      }
    }
  });

  if (changed) {
    savePillsToStorage();
    renderApp();
  }
}
```

5.3. Family Caregiver Profile Lifecycle & Cascade Deletion

Provides secure deletion of secondary family member profiles with comprehensive cascade removal of all associated medication plans and history logs, while protecting primary profile integrity:

```
// preview/index.html
function deleteProfile(profileId, profileName) {
  // Guardrail: Protected system accounts cannot be deleted
  if (profileId === 'all' || profileId === 'p1') {
    alert("Primary profile cannot be deleted.");
    return;
  }
  if (!confirm(`Are you sure you want to delete profile "${profileName}" and all associated medications?`

  // Cascade deletion across plans, history records, and profiles
  pills = pills.filter(p => p.profileId !== profileId);
  historyLogs = historyLogs.filter(h => h.profileId !== profileId);
  profiles = profiles.filter(p => p.id !== profileId);

  // Fallback to primary user if deleted profile was currently active
  if (activeProfileId === profileId) activeProfileId = 'p1';

  savePillsToStorage();
  saveHistoryToStorage();
  saveProfilesToStorage();
  renderApp();
}
```

5.4. Clinical Early Dose Guardrail Engine

To eliminate accidental double-dosing or premature administration (e.g., taking a night sedative or blood-pressure tablet hours ahead of time), the system intercepts confirmation clicks and validates the current timestamp against the scheduled regimen. If clicked more than 45 minutes ahead, execution halts with a protective clinical dialog:

```
// preview/index.html & lib/screens/home_screen.dart
function toggleTakenStatus(id) {
  const pill = pills.find(p => p.id === id);
  if (!pill) return;

  if (pill.isTaken === 0) {
    const now = new Date();
    const currentMins = now.getHours() * 60 + now.getMinutes();
    const [schedH, schedM] = pill.time.split(':').map(Number);
    const schedMins = schedH * 60 + schedM;
    const diffMins = schedMins - currentMins;

    // Guardrail: Block confirmation if more than 45 minutes early today
    if (diffMins > 45) {
      const hoursLeft = Math.floor(diffMins / 60);
      const minsLeft = diffMins % 60;
      alert(`⚠️ Too early to take medication! Scheduled for ${pill.time} (${hoursLeft}h ${minsLeft}m rem
      return;
    }

    // Advance cycle and synchronize group inventory...
  }
}
```

5.5. Card-Level In-Place Plan Editing (🖍️ Action) & Locked Formulation Tap Alerts

Provides an immediate mechanism to rectify incorrectly entered times or dosages. When editing an existing therapy card (e.g., correcting the scheduled hour of a night dose from 22:30 to 21:00), the system preserves clinical integrity by **locking the medicine formulation** (with a prominent `🔒 Type Locked in Edit Mode` badge). If the user attempts to tap any other formulation (e.g., attempting to change a Tablet plan to Drops), the system immediately intercepts the click, blocks the change, and displays an informative clinical modal alert (`⚠️ MEDICINE TYPE CANNOT BE CHANGED`) explaining that dosage integrity requires deleting and re-creating the plan if a formulation change is needed:

```
// preview/index.html & lib/screens/add_pill_screen.dart
function selectType(type, el) {
  if (editingPillId !== null && type !== selectedType) {
    alert("⚠️ MEDICINE TYPE CANNOT BE CHANGED\n\nThe formulation of an existing plan is locked to preser
    showToast("⚠️ Medicine type is locked. Add a new plan to change type.");
    return;
  }
  selectedType = type;
  // ...
}
```

```
// preview/index.html
function openEditMedicineModal(id) {
  const pill = pills.find(p => p.id === id);
  if (!pill) return;
  editingPillId = id;

  document.getElementById('inp-name').value = pill.name;
  document.getElementById('inp-stock').value = pill.stock;
  document.getElementById('inp-dosage').value = pill.dosage;
  selectType(pill.type || 'tablet');
  selectedFreq = pill.freq || 1;
  updateStepperUI();

  document.getElementById('add-modal-backdrop').classList.add('active');
}
```

5.6. Circadian Dose Slot Identification in Confirmation Alerts

To eliminate ambiguity when taking multi-dose medications (e.g., taking an evening dose vs. a morning dose of the same pharmaceutical), confirmation toasts and background notifications dynamically extract and display the specific circadian dose slot (`Sabah Öğünü` , `Öğle Öğünü` , `Akşam Öğünü` , `Gece Öğünü`):

```
// preview/index.html & lib/screens/home_screen.dart
const slotLabel = getPillSlotLabel(pill); // e.g. "Gece Öğünü" / "Night Dose"
const fullPillName = `${pName} (${slotLabel})`;

// Toast output: "İburamin Cold 2 (Gece Öğünü) alındı! Sonraki doz: 2 gün sonra."
showToast(lang.toastTakenDays(fullPillName, pill.freq));
```

5.7. Interactive Clinical Doctor Report & Family Member Breakdown

Rather than merely copying unformatted text to clipboard, the application provides an interactive clinical dialog. When "All Family" mode is active, historical doses are segmented patient-by-patient, displaying exact timestamps, formulations, circadian slots, and verified adherence rates:

```
// preview/index.html & lib/screens/home_screen.dart
function renderDoctorReportModalContent() {
  const isFamily = activeProfileId === 'all';
  // If family mode: groups records by family member (Abdullah, Can, Fatma)
  // Generates structured on-screen preview and monospace clinical export text
  // with verified 100% adherence certification and patient-specific dosage history.
}
```

5.8. Family Caregiver Clinical Demographics (Age, Gender, Weight, Height) & Strict Bilingual Localization

In clinical medicine, therapeutic dosages for pediatric and geriatric patients rely strictly on verified gender-specific pharmacokinetic clearance and body mass (mg/kg). To prevent medical ambiguity, the system enforces unambiguous binary gender classification (**Male** / **Female**) while keeping demographic data entry optional for frictionless onboarding. Furthermore, a comprehensive internationalization (i18n) engine guarantees zero untranslated residual words when operating in English mode:

In clinical medicine, therapeutic dosages for children and elderly patients frequently depend on body mass (mg/kg) and age-related organ clearance. The system allows caregivers to record optional clinical demographics (Age, Gender, Weight, Height) for each household member. These parameters are non-mandatory to maintain friction-free user onboarding, but when populated, they flow dynamically into the Clinical Doctor Report:

```
// preview/index.html & lib/models/pill_model.dart
const newProf = {
  id: 'p_' + Date.now(),
  name: nameInp,
  relation: relInp || 'Family Member',
  age: ageVal,       // e.g. 8 years
  gender: genderVal, // 'male' | 'female'
  weight: weightVal, // e.g. 26.0 kg (critical for pediatric mg/kg dosing)
  height: heightVal  // e.g. 128 cm
};
// Populates clinical demographics into Doctor Report:
// "Patient: Can (Child) • Age: 8 • Male • 26 kg • 128 cm"
```

6. QUALITY ASSURANCE & TEST VERIFICATION MATRIX

TEST CASE	INPUT SCENARIO	EXPECTED OUTCOME	RESULT
TC-01	User confirms daily Parol dose (Freq = 1)	Stock decrements by 1; entry appended to History Log	**PASSED**
TC-02	User confirms 2-day Devit-3 dose (Freq = 2)	Due date shifts exactly 2 days forward; removed from Today list	**PASSED**
TC-03	Stepper incremented to 4-day interval	Schedule frequency set to 4 days; helper text reflects 4-day period	**PASSED**
TC-04	User selects "Spray" formulation	"With food" option hidden; 5-Day decongestant warning shown	**PASSED**
TC-05	Language toggled to English (EN)	All labels, modal options, chips, and warnings translate to EN	**PASSED**
TC-06	Caregiver switches to "Can (Child)"	View isolates child's medicines; adherence recalculates for child	**PASSED**
TC-07	Inventory drops to 3 units	Low-stock audio chime and banner warning triggered	**PASSED**
TC-08	User schedules Parol 500 mg with Morning and Evening selected simultaneously	Generates two linked dose records with independent alarms (08:30 and 19:30) and unified `groupId`	**PASSED**
TC-09	User takes morning dose of linked Parol (stock: 40); next day arrives	Stock decrements to 39 across both morning and evening entries; at midnight `isTaken` auto-resets to 0 for next day's alarm	**PASSED**
TC-10	User deletes secondary profile "Can (Child)"	Prompts confirmation, cleans up all child medications and logs, and safely falls back to Primary Profile; delete button blocked for Primary Profile	**PASSED**
TC-11	User attempts to mark 22:30 night dose as taken at 14:00 (diff > 45 min)	Early Dose Guardrail blocks action, displays alert showing 8h 30m remaining, and preserves pending status	**PASSED**
TC-12	User taps 🖍️ edit icon on Parol card to correct scheduled time from 22:30 to 14:30	Opens modal pre-populated with active parameters; saving updates card time and re-arms alarm immediately	**PASSED**
TC-13	User creates Wound Dressing (Pansuman) and Topical Cream regimens	Meal selection automatically concealed; sterile pad & FTU presets rendered; sterility and mucosal precautions displayed	**PASSED**
TC-14	User taps 🖍️ edit icon on a card to correct its scheduled hour	Formulation grid is securely locked with badge "🔒 Locked in Edit Mode"; tapping other types displays toast refusal; hour update saves and synchronizes linked group stock	**PASSED**

TEST CASE	INPUT SCENARIO	EXPECTED OUTCOME	RESULT
TC-15	User confirms taking Iburamin Cold 2 night dose	Confirmation toast explicitly states exact circadian meal slot: "İburamin Cold 2 (Gece Öğünü) alındı! Sonraki doz: 2 gün sonra."	**PASSED**
TC-16	User clicks "View & Share Doctor Report" in All Family mode	Opens interactive modal displaying full clinical summary, segmented patient-by-patient (Abdullah, Can, Fatma) with times, formulations, dosages, and WhatsApp copy button	**PASSED**
TC-17	User in Edit Mode taps Drops chip on an existing Tablet plan	Click intercepted; system displays modal alert "Medicine Type Cannot Be Changed" with toast reminder; original formulation remains safely active	**PASSED**
TC-18	Caregiver edits "Can (Child)" profile, setting age: 8, weight: 26 kg, height: 128 cm	Demographics persist in local store, appear as tag on profile card, and flow into Clinical Doctor Report for physician dosage calculation	**PASSED**
TC-19	Caregiver switches UI to English and opens Family Profile form	Dropdown presents strictly "Male" and "Female" options; all profile fields, placeholders, clinical badges, and Doctor Report render in 100% English with zero Turkish residual characters	**PASSED**

7. FUTURE WORK & ACADEMIC CONCLUSION

Future development phases for **Medicine Reminder** include integrating automated PDF/WhatsApp export of 30-day adherence reports for clinical consultations, optical barcode/NFC packaging recognition for instant prescription entry, and drug-drug contraindication cross-referencing.

In conclusion, the project successfully bridges the gap between software engineering and clinical pharmacology. By implementing user-defined interval scheduling, circadian timing alignment, multi-profile caregiver segregation, clinical safety alerts, and a verifiable history journal, **Medicine Reminder** delivers a reliable, production-ready mobile health system.